MagicQ miniQ Wing

This handbook describes building your own lighting console modeled after the modern **MagicQ Compact Mini Connect layout**. The console controls the MagicQ software via **keyboard commands (HID emulation)** using the software's built-in playback shortcuts function.

> **Note:** The firmware uses a **5×13 button grid**. This handbook documents the keymap according to the **verified** state (checked via the firmware's diagnostic dump). The labeling of the individual keys can follow your own needs – what matters is the cell → key code mapping in the table below.

Table of contents

- 1. The layout (physical arrangement)
  - 1.1 Firmware keymap (verified)
  - 1.2 Fader calibration (no re-flashing needed)
- 2. Required hardware & shopping list
- 3. Hardware wiring
  - 3.1 The faders (via the 74HC4067 multiplexer)
  - 3.2 The key matrix (5x13 matrix) & diode wiring plan
- 4. Installation & flashing the bootloader
- 5. Setup in MagicQ
- Project structure

1. The layout (physical arrangement)

The console has **5 rows of 13 buttons** and the fader section below them.

- **Row 1 (GP2): DBO** + **NEXT PAGE** + the **S row** (select playback 1–10) with **REL** far right.
- **Row 2 (GP3): ADD/SWAP** + **PREV PAGE** + the **GO row** (GO 1–10) with **PAUSE MASTER** far right.
- **Row 3 (GP4): ALT** + the **flash row (F1–F10)** with **GO MASTER** far right.
- **Row 4 (GP5):** the **attribute keys** — GROUP, INT, POS, COL, BEAM, FX, SHIFT.
- **Row 5 (GP6):** free (unused cells, easy to extend).

```
                                        [ MINI CONNECT WING ]
=============================================================================
=========================
  DBO  | NEXT PAGE|  S1  |  S2  |  S3  |  S4  |  S5  |  S6  |  S7  |  S8  |
S9  |  S10 |  REL
  ADD/SWAP | PREV PAGE| GO1 |  GO2 |  GO3 |  GO4 |  GO5 |  GO6 |  GO7 |
GO8 |  GO9 |  GO10| PAUSE MASTER
  ALT  |    -     |  F1  |  F2  |  F3  |  F4  |  F5  |  F6  |  F7  |  F8  |
F9  |  F10 |  GO MASTER
=============================================================================
=========================

    ||                   ||      ||      ||      ||      ||      ||
||      ||      ||      ||
  Grand                [Fader] [Fader] [Fader] [Fader] [Fader] [Fader]
```

```
 [Fader] [Fader] [Fader] [Fader]
  Master                  1       2       3       4       5       6
7       8       9       10
```

1.1 Firmware keymap (verified)

The mapping is defined in `src/main.cpp` as `KEYMAP[5][13]` and checked via the diagnostic combination S10+GO10 (which types the cell table into the editor):

Cell	Key(s)	Key code	Function in MagicQ "Playback shortcuts"
0,0	DBO	`F11`	**Dead Black Out** (Mac keyboard layout)
0,1	NEXT PAGE	`[`	Next playback page
0,2–0,11	S1–S10	`1 2 3 4 5 6 7 8 9 0`	**Select Playback 1–10**
0,12	REL	`-`	**Release Playback**
1,0	ADD/SWAP	`` ` ``	**Add/Swap**
1,1	PREV PAGE	`]`	Previous playback page
1,2–1,11	GO1–GO10	`q w e r t y u i o p`	**GO Playback 1–10**
1,12	PAUSE MASTER	`#`	**Manual STOP**
2,0	ALT	`Ctrl+Alt+0` (hold)	**Dead Black Out (DBO)** short-circuit
2,1	–	–	unused
2,2–2,11	F1–F10	`\ z x c v b n m , .`	**Flash Playback 1–10** (momentary)
2,12	GO MASTER	`Space`	**Manual GO**
3,2	GROUP	`Ctrl+G`	Open **Group** window
3,3	INT	`Ctrl+I`	**Intensity** window
3,4	POS	`Ctrl+P`	**Position** window
3,5	COL	`Ctrl+K`	**Color** window
3,6	BEAM	`Ctrl+J`	**Beam** window
3,7	FX	`Ctrl+F`	**FX** window
3,8	SHIFT	`Shift` (held)	Modifier
4,0–4,12	–	–	free

Special firmware behaviors:

- **Flash (F1–F10):** MagicQ "Test" keys toggle the playback to 100% – the firmware therefore sends a key press (**on**) when pressed and again (**off**) when released. Result: a **momentary** flash while your finger is on the key.
- **S + GO (same playback) = PAUSE/STOP:** While S_n and GO_n are held together, the firmware sends the **STOP key** (`a s d f g h j k l ;`). STOP is a toggle in MagicQ – every press pauses/starts. To avoid a "ws" ghost key press, each S/GO key is delayed ~40 ms; the chord detection decides whether STOP or the individual key is sent.
- **Single keystroke (tap):** NEXT PAGE, PREV PAGE, REL, ADD/SWAP, GO MASTER, PAUSE MASTER and the window keys GROUP/INT/POS/COL/BEAM/FX send **one press+release** as soon as the button goes down — nothing is sent while held or on release.
- **Hold keys (ALT, SHIFT):** pressed like modifiers (`Ctrl+Alt+0`, `Shift`) until released. Held keys are re-typed every ~500 ms (keep-alive loop) because MagicQ's front-stop interrupts HID key presses mid-hold.
- **GO MASTER / PAUSE MASTER** sit on the **middle row** right (GO) / **row 2** right (PAUSE) instead of being stacked vertically.
- **Row 4 = attribute keys:** the graphical windows (GROUP/INT/POS/COL/BEAM/FX) and SHIFT now live on the wing (they used to be hardware buttons on the encoder module).

Fader handling (direct drag on the on-screen faders):

- The wing runs entirely in MagicQ **Playback shortcuts** keyboard mode (see section 5). The faders no longer send binary Test-key triggers. Instead they **grab and drag** the on-screen faders of the MagicQ **Full Panel** with the same absolute pointer mechanism as the sibling encoder wing:
  - A second HID device is registered — an **absolute mouse** (digitizer-style, `ordering 8`, so it precedes the keyboard reports; the pointer maps the whole screen to X=0..32767 / Y=0..32767).
  - Per fader the firmware jumps the cursor onto the fader track (button down), follows the physical fader, and releases again. Index 0 = **Grand Master**, 1..10 = **PB1..PB10**.
  - Output modes via `FADER_OUT_MODE`: **2 = DRAG** (current default), 1 = SET (keyboard entry of the value), 0 = WHEEL (calibrated wheel nudges).
- **Ghosting filter (Richtungs-Persistenz):** a read value is only accepted proportionally after the next scan either continues in the same direction or holds near the candidate target; genuine A→B→A flicker is discarded, so nothing is sent while real faders are idle. Large single-scan jumps (`FADER_MAX_DELTA_PCT`, 90) and a deadband (`FADER_DEADBAND`, 0) protect against artifacts.
- **Idle/Locked faders:** unwired multiplexer inputs float (settle-dependent readings: 30 µs → ~1 %, 3000 µs → mid, plus a ~2 s charge-up and drift). `FADER_IDLE_GUARD` marks a channel abnormal at boot when short-settle ≠ long-settle reading (`FADER_IDLE_SPREAD`), and `FADER_FORCE_LOCK` deterministically locks the unwired channels (currently f03–f07) — locked channels keep their reference but never emit anything.
- Faders are read with `FADER_SETTLE_US` = 1500 µs and averaged (`FADER_SAMPLES` = 12, every 5th scan skipped); both ends snap to exactly 0/100 within `FADER_ENDSTOP` (1 %).
- On boot a **fader ADC scan** prints every mux channel (`ch<no> raw=<adc> <pct>%`) to verify the 74HC4067 wiring.
- All knobs (mode, drag gain, sampling, locks) are tunable at the top of `src/main.cpp`.

1.2 Fader calibration (no re-flashing needed)

Restart the Pico while holding the DBO button → calibration mode starts automatically (4 LED blinks). The cursor is steered with the wing buttons, and **NEXT Page saves** the values permanently in the Pico's LittleFS flash (`/fcal.bin`).

Clear the stored calibration: restart the Pico while holding **REL**. The firmware deletes `/fcal.bin`, so the next boot uses the full-range defaults (useful after re-wiring, since stale endstop values would otherwise make every fader look `LOCKED`).

Key	Function
S1	X position of PB1
S2	X position of PB10
S3	Y of the PB track at 100 % (top) — cursor **on the knob**
S4	Y of the PB track at 0 % (bottom) — cursor **on the knob**
S5	X position of the Grand Master
S6	Y of the GM track at 100 % — cursor **on the knob**
S7	Y of the GM track at 0 % — cursor **on the knob**
F1 / F2	−500 / −50 (coarse / fine)
F3 / F4	+50 / +500
NEXT Page	save to flash and return to normal operation

PB2–PB9 are interpolated automatically between X(PB1) and X(PB10). Values are read back on every boot (`[CAL] flash …`, or `[CAL] default …` if no file exists yet). The compiled-in defaults of the current MagicQ Full Panel are: GM X = 400 (left of the Sub Master), PB X = 2800…25300 in steps of 2500, PB Y = 26800/29300, GM Y = 26850/29400.

Fader endstops (0 % / 100 %): A 10 k slide pot only covers part of the ADC range (tolerance, end caps, wiper end position), so without correction the top end stays below 100 % and the on-screen fader never reaches 100 %. Press **S8** to switch to the endstop mode:

Key	Function
S8	toggle between geometry mode and endstop mode
DBO / GO1… GO10	select the fader (DBO = Grand Master, GO1…GO10 = PB1…PB10)
S9 / S10	edit the lower (LO) or upper (HI) raw value of the selected fader
MASTER GO	take over the fader's currently measured raw value — push the fader fully down, press it for LO, fully up, press it again for HI
F1…F4	adjust the raw value manually (±500 / ±50)

`[RAW] … ist=… → …%` shows the live raw reading and the resulting percentage; a wrong endstop is immediately visible. Raw values are stored alongside the coordinates in `/fcal.bin`.

> **Wheel fallback (`FADER_OUT_MODE = 0`):** one wheel report = 0.40 % travel, globally paced to one report per 25 ms because MagicQ evaluates at most ~40 reports/s. Kept as a backup for setups without a Full Panel.

Diagnostics combos:

- **S10 + GO10** (keep both held): types the currently pressed matrix cells as `M R2C3 R0C1` into the focused editor.
- **S1 + GO1**: types the current fader levels as `F GM=42% PB1=87% ...`.
- **Boot mux diagnostic (`FADER_MUX_DIAG` in `src/main.cpp`, default off):** during `FADER_MUX_DIAG_MS` (35 s) after boot the firmware prints **all 16** multiplexer channels raw (`c00=… c15=…`), not just the wired ones. Move a fader in that window: a properly wired input sweeps the full range (~`0…65520`), a floating one drifts. This separates a **cold/dry solder joint on a mux input** from a firmware fault — a bad fader reads only a small, offset range (e.g. ~`2100…4900`) while a good fader on the **same** multiplexer spans the full `0…65520`.

2. Required hardware & shopping list

- **1x Raspberry Pi Pico** (standard or Pico H).
- **1x 74HC4067 multiplexer board** (16-channel analog multiplexer for reading the faders).
- **11x slide potentiometers (faders)**:
  - **Recommendation**: Bourns PTA6043 or ALPS RS6011Y.
  - **Specification**: **10k Ohm, LINEAR** (Important: labeled "B10K" or "10K Lin"). **60mm** travel corresponds to the original.
- **46x mechanical key switches** (5×13 grid, 44 cells populated + 2 spare):
  - **Recommendation**: Cherry MX Black or Cherry MX Red (linear feel without audible clicking for fast, silent flashing).
- **46x keycaps**: standard 1U size. Suggested color scheme: gray (S), blue (Flash), green (GO).
- **38x diodes**: model **1N4148** (indispensable for avoiding "ghosting" in the key matrix).

3. Hardware wiring

3.1 The faders (via the 74HC4067 multiplexer)

The Pico has only 3 analog inputs (ADC), but we need 11 inputs for 10 playbacks + 1 grand master. The multiplexer switches these extremely quickly onto a single pin.

- **Fader power**: connect the left pin of all 11 faders to **3V3(OUT)** (Pico pin 36) and the right pin of all faders to **GND** (e.g. Pico pin 38).
- **Fader signal lines**: the middle pin (wiper) of playbacks **PB1 to PB10** goes to inputs **C0 to C9** of the multiplexer. The grand master fader goes to input **C10**.
- **Multiplexer to Raspberry Pi Pico**:
  - **VCC** -> 3V3 (Pico pin 36)
  - **GND** -> GND (Pico pin 38)
  - **SIG / COM** -> **GP27** (Pico pin 32 / ADC1)
  - **S0** -> **GP20** (Pico pin 26)
  - **S1** -> **GP21** (Pico pin 27)
  - **S2** -> **GP22** (Pico pin 29)

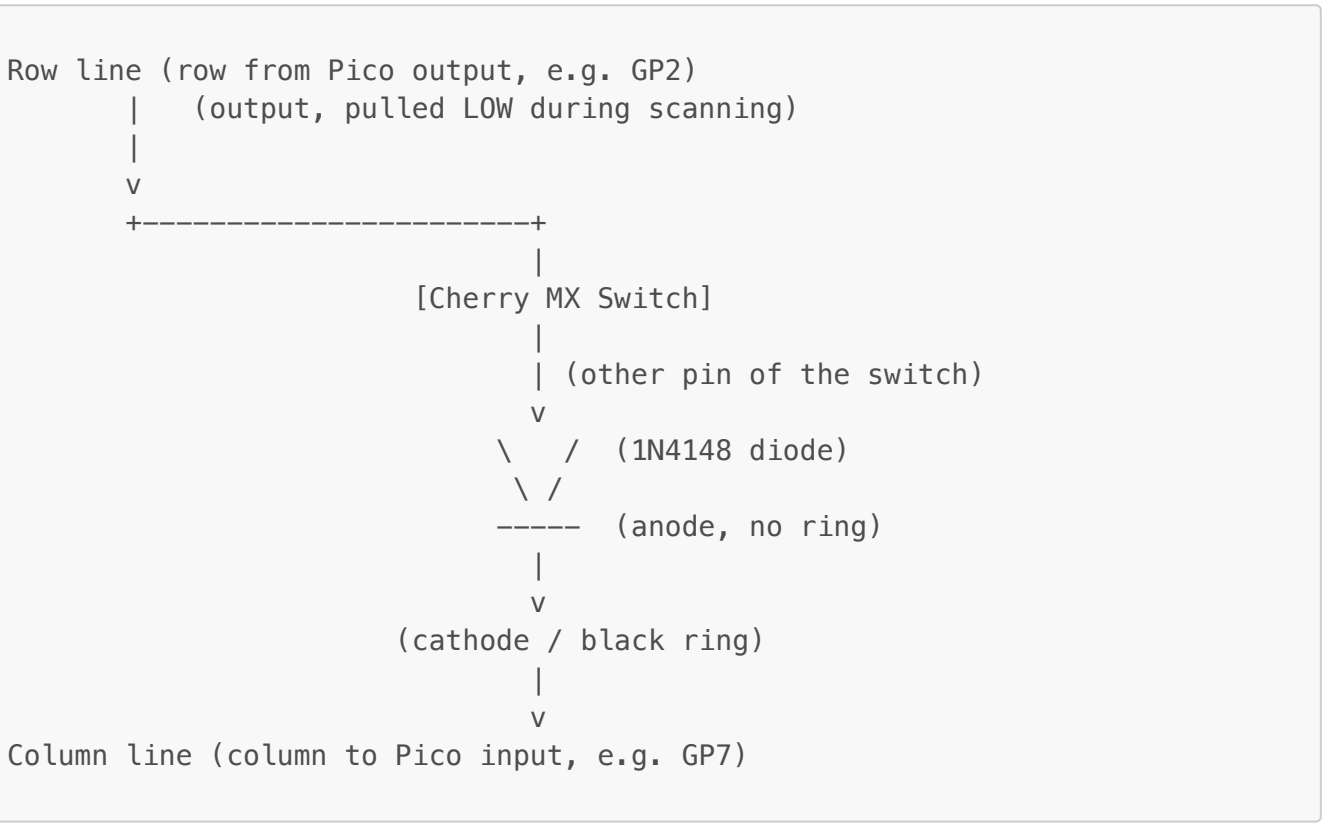
- **S3** -> **GP26** (Pico pin 31 / ADC0)

3.2 The key matrix (5x13 matrix) & diode wiring plan

To save pins, we wire the keys in **5 rows and 13 columns**. A **1N4148** diode is soldered onto each switch to prevent false triggers when pressing multiple keys at once (ghosting).

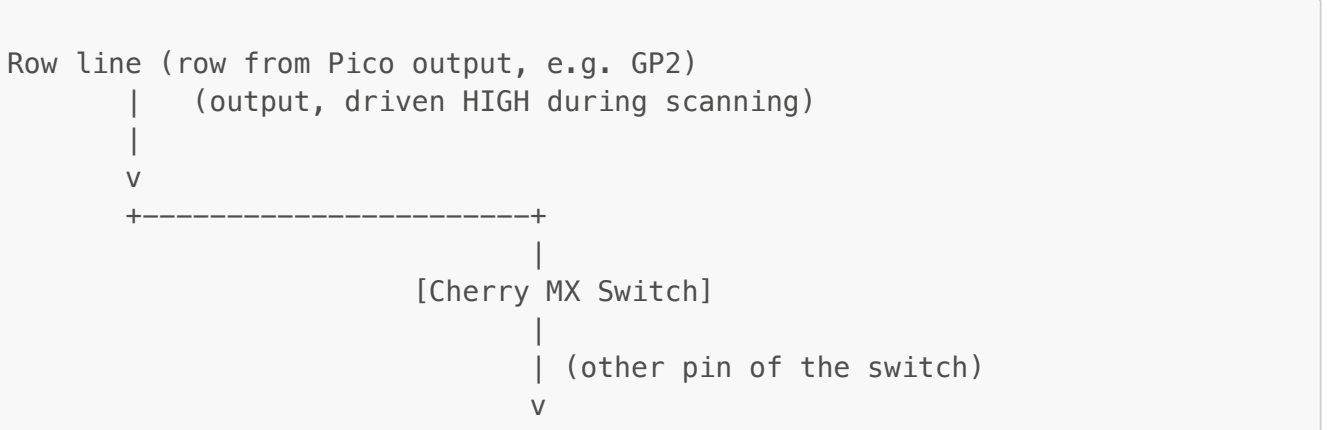
There are two valid combinations of diode direction and scan polarity — the sections must match!

Variant A "Active-Low" (original, diode cathode toward row): Diode: anode toward column, **cathode (black ring) toward row/switch**. Scanning: rows pulled LOW, columns with internal pull-ups, key detected at LOW.



Current path when pressing a key: **Column** (pull-up, HIGH) -> **Anode** -> **Cathode** -> **Switch** -> **Row** (LOW). The diode conducts current from the column to the low row and pulls the column LOW.

Variant B "Active-High" (standard on many PCBs, diode cathode toward column): Diode: **cathode (black ring) toward column**, anode toward row/switch. Scanning: rows driven HIGH, columns with internal pull-downs, key detected at HIGH.



```
                                \   /  (1N4148 diode)
                                 \ /
                                 -----  (cathode / black ring)
                                   |
                                   |
                                   v
Column line (column to Pico input, e.g. GP7)
```

Current path when pressing a key: **Row** (HIGH) -> **Switch** -> **Anode** -> **Cathode** -> **Column** (pull-down, LOW). The diode conducts current from the high row to the column and pulls the column HIGH.

Caution with existing wiring: If the diodes are already soldered and point toward the **Column** (original handbook), you do **not** need to re-solder them — the firmware (`src/main.cpp`) scans **Active-High** and works with that. The variant described above (cathode toward row) is what a fresh board expects.

Pin assignment on the Pico for the matrix:

- **Rows (outputs)**: Row 0 (DBO/S/REL) = **GP2**, Row 1 (ADD/SWAP + GO/PAUSE) = **GP3**, Row 2 (ALT + F1–F10 + GO MASTER) = **GP4**, Row 3 (attributes) = **GP5**, Row 4 (free) = **GP6**
- **Columns (inputs)**: Column 1 to 13 on **GP7 to GP19**

4. Installation & flashing the bootloader

The firmware is **latency-optimized** C++ (compiled to native ARM code) and uses the built-in USB-HID keyboard of the **earlephilhower Arduino core** — the same approach as the sister project `Enncoder`. It is built with **PlatformIO** (`platformio.ini` in the project root).

```
# Open and build the project:
cd Wing
pio run           # builds firmware.uf2

# Flash to the Pico:
pio run -t upload
```

Uploads use the RP2040's native **1200-bps CDC touch** (core `SerialUSB` + `use_1200bps_touch` in the `rpipico` board): `pio run -t upload` reboots the running device into BOOTSEL by itself, so **no BOOTSEL button** is needed after the first flash. Only the very first flash of the Pico requires **BOOTSEL** (hold while plugging in USB, then release).

`-DENABLE_PICOTOOL_USB` must **not** be set: it makes the running firmware advertise the Raspberry Pi USB reset interface, but on this composite HID stack it broke USB enumeration on macOS (no `/dev/cu.usb*` port, no HID keyboard). Pin the touch port via `upload_port` (the `usbmodem` name can change after replugging — check with `pio device list`).

`board_build.filesystem_size = 64k` reserves the 64 KB **LittleFS** partition that `/fcal.bin` needs (a bare `64` would mean 64 *bytes*, not 64 KB, and hangs the boot with interrupts disabled — always use the `k`).

The generated file is at `.pio/build/pico/firmware.uf2` (alternatively by drag & drop onto the `RPI-RP2` drive after starting the Pico in BOOTSEL mode).

5. Setup in MagicQ

1. Start MagicQ, select **Setup** -> **View Settings**.
2. Switch to the **Keypad Encoders** tab.
3. Find the **MagicQ PC Keyboard Mode** line and change it to **Playback shortcuts**.
4. Ideally set your PC's operating system to the **US keyboard layout** for the DIY keyboard, so that special characters (`#`, `` ` ``, `\`, `,`) used by the wing are typed correctly by the Pico.
5. **Faders:** open the **Full Panel** window so that all 10 playback faders and the Grand Master are visible — the wing drags them there. If MagicQ shows a different window, open the Full Panel once before using the faders (the coordinates are calibrated for that layout).
6. **macOS tip**: If the Mac's physical Shift key occasionally drops out while the Pico is connected as an additional USB keyboard — that is known macOS behavior with multiple HID keyboards (the Pico injects Shift for characters like `#`). The firmware therefore sends lowercase letters (MagicQ evaluates key codes case-insensitively). If you don't want that at all, unplug the Pico when typing only on the Mac keyboard.
7. **Linux/Windows note**: The flash keys (F1–F10) work on the Mac, **not** on the PC (MagicQ-PC has no F-key binding in "Playback shortcuts"). The firmware therefore uses the **Test keys** (`\ z x c v b n m , .`) instead, which work on every system and toggle (100% playback on/off). The firmware automatically compensates the toggling into **momentary flash**.

Project structure

```
Wing/
├── README.md            # This handbook
├── platformio.ini       # PlatformIO configuration (earlephilhower core)
├── pcb/                 # Gerbers, BOM and pick&place for the wing PCBs
└── src/
    └── main.cpp         # C++ source (latency-optimized, USB-HID)
```